

CDMP User Guide

CDMP (Canadian Donation Map) is a web application that makes three decades of Canadian federal political donation data (1993–2024) explorable through an interactive, geography-first map. Anyone can open the app and explore aggregated donation patterns by party, year, and region without creating an account. University researchers can additionally log in to a secure research dashboard to search, filter, and export individual-level donation records for academic study. CDMP also includes **Ask CDMP**, an AI assistant that answers plain-language questions about the aggregate donation data. This guide supplements the introductory walkthrough video, explains how to run the product, and describes what each feature does.

Introductory Video

 CDMP_IntroductionVideo_TeamNACSS.mp4

Getting Started: Downloading the Data and Running the Product

CDMP runs locally with Docker. The application code lives in the GitHub repository, and the large data artifacts (a restorable MongoDB dump plus the raw source data) are shared separately through Google Drive because they are too large for GitHub.

- DOCKER.md — Docker setup, MongoDB dump restore, and full rebuild steps
- DATA_README.md — the expected layout of the external data package
- README.md — project overview, AI provider options, and data-coverage notes

Prerequisites

- Docker Desktop, installed and running.
- The data package ZIP shared via Google Drive. It contains the MongoDB dump and the raw data.
 - <https://drive.google.com/file/d/1xQCDA2nmejBIbpYF7N54gwZZ8EDKn5G/view?usp=sharing>
- (Optional) An AI provider API key if you want to use Ask CDMP, OpenAI-compatible, Anthropic Claude, or Google Gemini. The map and research features run without one; only Ask CDMP requires a key.

Run it (recommended: MongoDB dump restore)

1. Install and open Docker Desktop.
2. Clone the repository, then move into it:

```
git clone https://github.com/UTSC-CSCC01-Software-Engineering-I/course-project-nacss.git
cd course-project-nacss
```

3. Download the data ZIP from the Google Drive link:<https://drive.google.com/file/d/1xQCDA2nmejBIbpYF7N54gwZZ8EDKn5G/view?usp=sharing> into the project root and unzip it:

```
unzip -n course-project-nacss-data-drive-upload.zip
```

After unzipping, the file `mongo-dump/course_project_nacss.archive` should exist in the project.

4. (Optional, for Ask CDMP) Create the backend env file and add a provider key:

```
cp server/.env.example server/.env
```

- a. Open server/.env
- b. Replace JWT_SECRET, by running the following in your terminal and copying the output as the JWT_SECRET
 - i. openssl rand -hex 32
- c. Set AI_PROVIDER, AI_MODEL, and AI_API_KEY for one provider (e.g. OpenAI gpt-4o-mini, Claude claude-sonnet-4-6, or Gemini gemini-3.5-flash-lite). Only one provider is active at a time.
 - i. For more information, research online how to obtain an API key for the given AI of your choice.

5. Start the containers:

```
docker compose up --build -d
```

6. Restore the MongoDB dump into the database container:

```
docker compose run --rm mongo-tools mongorestore --host mongo --drop  
--archive=/dump/course_project_nacss.archive
```

7. (Optional) Confirm the restore loaded the expected data:

```
docker compose --profile tools run --rm data-importer npm run check:data
```

You should see roughly 5.37 million donation records and matching region-stat and riding-assignment counts.

8. Open the app in your browser:

```
http://localhost:8080
```

The backend health check is at <http://localhost:5001/api/health>.

9. When you are finished, stop the app:

```
docker compose down
```

Note: A fresh database needs the dump restore above before the map shows data. The map and research features work without an AI key; Ask CDMP will simply report that it is unavailable until a provider key is configured. To instead rebuild the database from raw source files end-to-end, follow the “Full Data Rebuild” path in DOCKER.md.

Creating a research account

To view individual-level records, click **Research Login** and then **Sign Up**, register with a university email address (ending in .ca or .edu), and choose a password. Log back in with those credentials and accept the privacy agreement once when prompted.

How to Use CDMP

Exploring the public donation map

- On arrival to the website, the map shows total national donations, broken down by province. No account is needed.
- Click any province to drill into a riding-level map for that province. Riding boundaries are shown for the historical boundary set that matches the currently selected time period.
- Click a riding to open its summary panel: total donations, donor count, party breakdown, and a longitudinal trend line for that riding.

- Use the search bar to jump straight to a riding by name (available at both the national and riding-level views).
- Click **Back to National View** (top-left) to return from a riding to its province, or from a province to the national view.

Filtering the map

Open the **Filters** panel to narrow the map by:

- Party (Liberal, Conservative, NDP, Bloc Québécois, Green, People's, or All Parties)
- Year (a specific year, or all years)
- Donor type (individuals, organizations, or all)
- Boundary-vintage bucket (available once you drill into a province): ridings changed shape over time, so this selects which historical riding boundaries to draw and which years of data to show against them
- Donations Count view, which adjusts totals by donation count so regions of different sizes can be compared fairly

If a selected combination of filters has little or no data for a region, the app explains this instead of showing a blank or broken view.

Accessing individual-level research data

1. Register via **Research Login** → **Sign Up** using a university email (ending in .ca or .edu) and a password.
2. Log in via **Research Login** and enter your credentials.
3. Read and accept the privacy agreement. This is required once before any individual-level record is shown, and it is logged.
4. On the research dashboard, you can:
 - Filter individual donation records by donor type, province, party, year, riding, and donor name (search)
 - Page through the filtered results in a table
 - Review the analytics panel for your current filters (see Main Features)
 - Click **Export** to download the currently filtered records as a CSV file
 - Go to **My Account** to view account info or change your password

Every login, query, and export is written to an activity log for audit purposes, in line with the platform's data-access policy.

Main Features

Interactive national / province / riding map The core map lets any visitor drill from a national view down to a province, then down to a riding, with boundaries redrawn per the historical boundary-vintage bucket that matches the selected time range.

Filters (party, year, donor type, boundary bucket) All filters apply live to the map and update totals, party breakdowns, and trend lines without a page reload.

Riding / region summary panel Clicking a region opens a panel with total donations, per-capita figures, donor counts, party breakdown, and a trend line over time for that region.

Search A search bar lets users jump to a riding by name, scoped to the current province once drilled in.

Authenticated research dashboard University-affiliated users can log in to browse individual donation records (donor type, amount, date, party, riding) behind a privacy agreement and authentication gate.

CSV export From the research dashboard, the currently filtered set of individual records can be exported as a CSV file for offline analysis.

Research analytics panel Alongside the filtered records table, the dashboard shows supplementary charts for the current filter set, top ridings by donation count and the distribution of donation amounts (donation concentration), so researchers get quick summary statistics without leaving the dashboard.

Account management Authenticated users can view their account details and change their password from the Account page.

Activity logging Every login, filtered query, and CSV export by an authenticated user is recorded to an activity log, supporting the platform's audit and privacy commitments.

Ask CDMP (AI assistant) A conversational assistant that answers plain-language questions about the aggregate donation data — totals, rankings, trends, comparisons, and year-over-year changes — and can apply its answer straight back onto the map. See the dedicated section below.

Using Ask CDMP (the AI Assistant)

Ask CDMP lets you explore the donation data by asking natural-language questions instead of manually selecting every filter. It is designed for students, researchers, journalists, and the public who want to investigate political-finance patterns without knowing the database structure. It answers *only* from the aggregate donation data inside CDMP, it is not a general-purpose political chatbot.

Where to find it and how to ask

- Open the map page. The **Ask CDMP** panel appears in the side panel as you scroll to the bottom, titled “Ask CDMP” with the prompt “Ask a question about Canadian political donations.”
- Type a question in the “Ask a question...” box and submit. The panel shows a **CDMP answer** card with the result, the filters it interpreted, and a data-coverage note.
- Use the **Try asking about this view** example prompts to get started, and the **Follow-up questions** suggestions to keep exploring.
- When a result maps cleanly onto the map, an **Apply to map** button lets you push it into the map filters. **Start over** clears the conversation.
- Requires a configured AI provider key (see setup above). Without one, Ask CDMP reports that the AI service is unavailable.

What it can answer

Question types: a single aggregate value; a ranked list of parties, provinces, or ridings within one province; a yearly trend; comparisons between parties, provinces, or two years; the change between two years; and a party breakdown within a named riding.

Metrics: total donation amount, number of donations, number of donors, average donation amount, and per-capita donation amount (where population data is available).

Example questions

- “How much was donated nationally in 2023?”
- “Which party received the most donations in 2023?”
- “Which five provinces received the most donations in 2023?”
- “Show the top ridings in Ontario.”
- “Show the Liberal donation trend in Ontario from 2019 to 2023.”
- “Compare Liberal and Conservative donations nationally in 2023.”
- “Which party increased donations the most from 2019 to 2023?”
- “Rank parties by donations in Ajax.”

Tips for riding names and time periods

- Use an official riding name (e.g. “Ajax,” “Davenport,” “Toronto Centre,” “Kitchener—Waterloo”). Capitalization, accents, spaces, and common punctuation are normalized, so “Kitchener-Waterloo” still matches “Kitchener—Waterloo.”
- Informal areas are not understood, e.g. “the GTA,” “downtown Toronto,” “Northern Ontario,” “my local riding.” If a name is ambiguous, add the province (e.g. “Victoria, British Columbia”).
- Riding questions must stay within one boundary period; 1997–2003, 2004–2014, or 2015–2024, because boundaries were redrawn between them. “Show the Davenport trend from 2015 to 2023” works; “...from 2010 to 2023” crosses two periods and is not combined.
- When no year is given, defaults apply: totals, rankings, and comparisons use 2023; trends and changes use 2019–2023 (2023 is the latest broadly comparable year in the dataset).

Safety, privacy, and limits

- Ask CDMP returns aggregate results only. It never returns donor names, postal codes, exact individual records, or contact information, and it suppresses results with very small donor counts under the project’s privacy threshold.
- The AI only interprets your question into a restricted, validated query; the application controls what data can be queried, how it is aggregated, and what can be returned. Unsafe or out-of-scope requests are rejected.
- It does not do arbitrary math, percent-change, forecasting, correlation/regression, demographic analysis, or questions about elections, polling, candidates, or policy. It cannot rank every riding in Canada at once, compare two named ridings directly, or infer your location.
- Data coverage is 1993–2024 but is not complete for every party/year: Conservative records are missing for 2020, Liberal records for 2021, and the 2024 data currently contains only Bloc Québécois records; riding data begins in 1997. Ask CDMP shows a coverage note when your selected period includes a known gap.
- Provider support: Ask CDMP works with OpenAI-compatible models, Anthropic Claude, or Google Gemini through a single adapter; the active provider is set in the backend environment.

Handled Edge Cases

CDMP degrades gracefully rather than showing a blank screen or crashing:

- Clear messaging when the map or dashboard data fails to load.

- A no-data explanation when riding boundaries/stats haven't been built yet for a selected time range.
- A no-data banner for the 2025-onward boundary bucket (donation data is not yet available past 2024).
- "No Regions Found" / "No Ridings Found" messaging for searches with no matches.
- "No records found for the current filters" in the research table.
- Inline errors if records fail to load, CSV export fails, or login fails.
- An Access Denied page for non-researcher accounts attempting research routes.
- A prompt to accept the privacy agreement before continuing, if not yet accepted.
- For Ask CDMP: a clear message for unsupported questions, and "try again" messaging if the AI service is busy or temporarily unavailable.

All of these use plain, non-technical language and never block navigation, filters, or search elsewhere in the app.